Supporting Information

The quantitative currency of GPCR bias signalling fails its own algebraic identities

Huan Li
Shanghai Liangta Biotech Co., Ltd., Shanghai 201899, P. R. China

October 2026

This Supporting Information documents the complete computational output of the accompanying repository (`gpcr-field-audit`): every analysis script, the adjudication tables, the per-source meta-audit table, and the provenance of third-party data. Numbers are transcribed verbatim from `results/`; scripts are plain Python 3 (numpy/scipy/matplotlib/pandas). The thirteen analysis scripts run in manuscript order under `run_all.py`, which archives each console output to `results/logs/` and writes a manifest recording package versions, exit codes and file hashes; `PAPER_MAP.md` maps every script to the panel or table it backs.

Contents

1 Pipeline map: scripts and outputs **2**

2 Atlas meta-audit (code67/71) **2**

3 MOR extended-model residuals (code80) **5**

4 Third-party data provenance **5**

1 Pipeline map: scripts and outputs

§	Script	Computes	Archived output
§3.1	code63	D2R binding-versus-function fracture (Fig. 1a, 2b)	results/logs/
§3.1	code64	D2R upstream rate ordering	results/logs/
§3.1	code68	D2R kinetic occupancy model (Fig. 2c)	results/logs/
§3.1	code69	D2R cross-arm spread versus time (Fig. 2a,c)	results/logs/
§3.2	code65	AT1R cross-arm affinity fracture (Fig. 1b, 2d)	results/logs/
§3.3	code66	μOR cell-level identity map (Fig. 1c, 2e)	results/logs/
§3.3	code80	μOR one-coordinate-per-arm model (Fig. 1c, 2f)	code80_*.csv, five files
§3.4	code67	Atlas meta-audit and pair deviations	results/logs/; atlas_pair_violations.csv under --write
§3.4	code71	Atlas identity pipeline; entry, group and paper tables	atlas_audit/*.csv, atlas_audit_summary.json
§3.5	code73	common-mode error, aggregation, jackknife	atlas_confidence_matrix.csv
§3.6	code77	constraint projection, fracture probability	atlas_repair_projection.csv, atlas_fracture_probability.csv
§4	code72	false-positive rate, power, controls	results/logs/
§3.4	code72b	fixed-cut independent-error null (entry and pair level)	results/logs/

Rows are in manuscript order. Full script names, the reproduction command and the caveats (pandas grouping; which numbers are console-only) are in PAPER_MAP.md.

Two scripts in the repository, code47 and code47b, implement a pilot adjudication of a boundary predictor on an unrelated calcium-transient dataset (Keshelava 2018). They are kept with their outputs under results/ for traceability but are not part of this manuscript and are not summarised here.

2 Atlas meta-audit (code67/71)

Per-source median absolute deviation from the algebraic identities, in dex, over all receptor/pathway groups in the BiasedSignalingAtlas community tables (Table 2). The full pair-level violation table (all receptor×pathway×ligand comparisons) is archived as atlas_pair_violations.csv (2.8 MB).

Table 2: Per-source-paper median and maximal identity deviation (dex).

DOI	n_{grp}	n_{pairs}	median	max	receptors
10.1111/bph.13854	2	4	1.322	2.209	OPRK_HUMAN
10.1124/jpet.116.232561	2	226	1.077	4.071	CNR2_MOUSE
10.1124/mol.115.101980	4	50	0.982	4.990	CNR1_HUMAN
10.1124/mol.113.088880	5	50	0.827	3.676	ADRB2_HUMAN
10.1074/jbc.M116.757559	7	13	0.682	1.966	CCR2/CCR5_HUMAN
10.1073/pnas.1312515110	3	3	0.676	1.357	CXCR4_HUMAN
10.1073/pnas.1908662116	4	28	0.531	1.496	OPRM_MOUSE
10.3389/fphar.2020.00188	1	1	0.529	0.529	OPRK_HUMAN
10.1016/j.bcp.2018.04.021	8	48	0.526	1.843	AGTR1_HUMAN
10.1073/pnas.2000712117	6	1240	0.452	4.569	OPRD/OPRK/OPRM_HUMAN
10.1007/s00018-013-1461-7	2	16	0.429	0.722	OPRD_HUMAN
10.1111/bph.14913	5	214	0.418	4.556	OPRD/OPRK/OPRM (H+M)
10.1124/mol.110.068106	2	57	0.400	1.796	DRD2_HUMAN
10.1016/j.bmc.2017.08.037	2	91	0.355	1.654	DRD2_HUMAN
10.1038/ncomms13958	8	288	0.326	4.054	CNR2_HUMAN/MOUSE
10.1074/jbc.M115.704122	1	15	0.287	0.610	NPSR1_HUMAN
10.1074/jbc.M112.408534	3	18	0.283	0.614	ACM2_HUMAN
10.1038/s41598-017-15258-z	2	13	0.270	0.664	OPRM_HUMAN
10.1124/mol.115.099192	2	42	0.267	0.737	CNR1_HUMAN
10.1016/j.phrs.2017.08.008	3	15	0.263	0.765	CNR1_HUMAN
10.1016/j.phrs.2018.02.032	5	225	0.262	1.275	APJ_HUMAN
10.1021/acs.jmedchem.8b00047	1	136	0.255	0.844	AA1R_HUMAN
10.1038/nchembio.2527	2	13	0.220	0.342	DRD2_HUMAN
10.3390/molecules24244542	4	351	0.210	0.847	OPRD/OPRM (H+M)
10.1124/mol.115.099150	3	27	0.205	1.822	OPRX_HUMAN
10.1002/pep2.24070	4	288	0.196	1.714	OPRD_MOUSE/OPRM_HUMAN
10.1038/s41467-018-03941-2	2	132	0.194	1.480	GLP1R_HUMAN
10.3390/ijms20102417	5	46	0.183	0.560	CCR1_HUMAN
10.1126/scisignal.aaz3140	5	157	0.178	1.452	OPRM_HUMAN/MOUSE
10.1124/jpet.114.220293	2	110	0.178	0.652	DRD2_HUMAN
10.1016/j.lfs.2016.02.069	1	6	0.167	0.328	EDNRA_HUMAN
10.1016/j.bcp.2014.05.008	3	15	0.163	0.728	AA2BR_HUMAN/MOUSE

DOI	n_{grp}	n_{pairs}	median	max	receptors
10.1038/s41467-019-11875-6	30	5399	0.162	3.325	ADRB2/OPRD/OPRM (H+RAT)
10.1016/j.euroneuro.2018.12.013	3	13	0.160	0.329	OPRD_HUMAN
10.1021/acschemneuro.9b00429	5	20	0.157	0.316	CNR1_HUMAN
10.1038/s41598-017-01078-8	2	2	0.153	0.249	LSHR_HUMAN
10.1038/s41598-020-65636-3	7	73	0.148	1.809	ADRB1_HUMAN
10.1016/j.bcp.2015.11.013	2	30	0.147	1.300	AA1R_HUMAN
10.1124/mol.115.098848	6	562	0.138	3.363	OPRM_HUMAN
10.1124/mol.113.089649	2	930	0.137	0.739	OPRK_HUMAN
10.1161/ HYPERTENSIONAHA.114.05099	3	3	0.132	0.762	APJ_HUMAN
10.1021/jm500457x	2	49	0.120	0.417	DRD2_HUMAN
10.1126/scisignal.aar4309	4	12	0.119	0.986	OPRK_HUMAN
10.1007/s00210-014-1054-5	5	417	0.117	0.568	ADRB2_HUMAN
10.1021/acs.jmedchem.8b01912	4	18	0.116	0.265	FPR1/FPR2_HUMAN
10.1124/mol.116.103283	2	210	0.115	0.588	AA3R_HUMAN
10.1124/mol.116.106153	4	40	0.112	0.688	HRH3_HUMAN
10.1021/acs.jmedchem.7b00363	2	110	0.110	0.745	DRD2_HUMAN
10.1126/science.1232808	4	4	0.109	0.331	5HT1B/5HT2B_ HUMAN
10.1124/mol.110.067454	1	21	0.107	0.273	ADA1A_HUMAN
10.1124/mol.110.065664	2	30	0.102	0.335	GLP1R_HUMAN
10.1126/scisignal.aat1631	8	524	0.101	1.532	AGTR1_HUMAN
10.1021/acsptsci.0c00022	3	217	0.100	0.693	GLP1R_HUMAN
10.1074/jbc.M113.504381	2	49	0.096	0.250	OPRK_HUMAN
10.3389/fphar.2018.01202	3	45	0.083	0.454	CNR2_HUMAN
10.1021/acs.jmedchem.6b01857	2	141	0.081	0.340	DRD2_HUMAN
10.1021/acs.jmedchem.6b01875	2	240	0.080	0.803	DRD2_HUMAN
10.1038/s41401-018-0200-x	2	342	0.078	0.316	ADRB2_HUMAN
10.1074/jbc.M115.659250	8	17	0.074	0.823	GHSR_HUMAN
10.1016/j.bcp.2017.03.014	4	12	0.067	0.190	AA2BR_HUMAN
10.1016/j.bcp.2016.05.014	6	168	0.064	0.818	OPRM_ HUMAN/MOUSE
10.1016/j.neuropharm.2019.107718	4	36	0.063	0.177	OPRM_HUMAN
10.1021/acsptsci.9b00083	2	342	0.050	0.327	CALRL_HUMAN
10.1016/j.bcp.2017.03.018	4	147	0.042	0.304	GLP1R_HUMAN
10.1021/jm401318w	2	218	0.040	0.296	DRD2_HUMAN
10.1101/2020.11.09.375162	18	3608	0.001	0.364	CNR1/CNR2_HUMAN

3 MOR extended-model residuals (code80)

The Gillis arm-coordinate extended model is fitted to the MOR ligand series; per-(nanobody, ligand, channel) residuals with standard deviations, model-1 residuals and arm shifts are archived verbatim in `mor_code80_residuals.csv` (67 rows). Example rows: Nb33 / Morphine: $\log \tau$ residual $+0.031 \pm 0.102$, $\log R$ residual -0.021 ± 0.092; Nb33 / Oxycodone: $\log \tau$ $+0.192 \pm 0.146$.

4 Third-party data provenance

Shipped here: `keshelava2018/peak_values.csv` (Supplementary Data 3 of Keshelava et al. 2018, CC BY 4.0; authors' MATLAB script carries a legacy dose ladder — our usage follows the Methods ladder, documented in the manuscript Methods). Referenced but not re-hosted: BiasedSignalingAtlas community tables (code67/71/77); Gillis et al., *Sci. Signal.* **13**, eaaz3140 (2020) main Tables 1–3 and SI Table S1 ($\sim$98 MB, not mirrored); D2R/AT1R source studies cited in the manuscript (Klein Herenbrink et al. 2016; Wingler et al. 2020). Full notes in `data_external/PROVENANCE.md`.